← Back **Foundations of Deep Learning - Assessment**
Graded Assignment • 20 min

Due Mar 4, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

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1. What is the mathematical formula for the ReLU activation function?

- (●) f(x) = max(0, x)
- ( ) f(x) = 1 / (1 + e^-x)
- ( ) f(x) = tanh(x)
- ( ) f(x) = x

Correct
This is Correct. The ReLU function outputs the input directly if it is positive, otherwise, it outputs zero. For more information, refer to the video "Training a Neuron-Demo".

1 / 1 point

2. What is the purpose of applying an activation function in a neuron?

- ( ) To normalize the input values
- (●) To introduce non-linearity into the model
- ( ) To calculate the weighted sum of inputs
- ( ) To produce the final prediction

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Correct
This is Correct. Activation functions introduce non-linearity, allowing the network to learn complex patterns in the data. For more information, refer to the video "Deep Learning Neural Network - Forward Propagation".

1 / 1 point

3. In the equation Z = W * X + b, what does Z represent?

- ( ) The weight matrix
- ( ) The input matrix
- ( ) The bias vector
- (●) The matrix of weighted sums for all neurons in a layer

Correct
This is Correct. Z is the result of the matrix multiplication between the weight matrix (W) and the input matrix (X), plus the bias vector (b). It holds the weighted sums for all neurons in a layer before the activation function is applied. For more information, refer to the video "Deep Learning Neural Network - Forward Propagation".

4. What is the primary goal of backpropagation in neural networks?

- ○ To initialize the model's parameters
- ○ To make predictions on new data
- ◉ To minimize the overall error and improve model accuracy
- ○ To introduce non-linearity into the model

Correct
This is Correct. Backpropagation's core purpose is to iteratively adjust the network's weights and biases to reduce the error between its predictions and the true values, thereby enhancing the model's accuracy. For more information, refer to the video "Backward Propagation - Deep Learning Neural Network".

1 / 1 point

5. Which step involves feeding the input data through the network to generate a prediction?

- ◉ Forward Pass
- ○ Loss Calculation
- ○ Backward Pass
- ○ Weight Initialization

Correct
This is Correct. The Forward Pass is where the input data travels through the network, layer by layer, resulting in the final output or prediction. For more information, refer to the video "Backward Propagation - Deep Learning Neural Network".

1 / 1 point

6. Given a categorical feature with values ['red', 'green', 'blue'], what would be the one-hot encoded representation of 'green'?

- ○ [1, 0, 0]
- ◉ [0, 1, 0]
- ○ [0, 0, 1]
- ○ [1, 1, 0]

Correct
This is Correct. In one-hot encoding, each category gets its own binary column. 'Green' is the second category, so its representation is [0, 1, 0]. For more information, refer to the video "Deep Learning, Training a Multi Class Classifier-Demo".